

GuideCopter - A Precise Drone-Based Haptic Guidance Interface for Blind or Visually Impaired People

Felix Huppert
University of Passau
Passau, Germany
felix.huppert@uni-passau.de

Gerold Hoelzl
University of Passau
Passau, Germany
gerold.hoelzl@uni-passau.de

Matthias Kranz
University of Passau
Passau, Germany
matthias.kranz@uni-passau.de

ABSTRACT

Drone assisted navigation aids for supporting walking activities of visually impaired have been established in related work but fine-point object grasping tasks and the object localization in unknown environments still presents an open and complex challenge. We present a drone-based interface that provides fine-grain haptic feedback and thus physically guides them in hand-object localization tasks in unknown surroundings. Our research is built around community groups of blind or visually impaired (BVI) people, which provide in-depth insights during the development process and serve later as study participants. A pilot study infers users' sensibility to applied guiding stimuli forces and the different human-drone tether interfacing possibilities. In a comparative follow-up study, we show that our drone-based approach achieves greater accuracy compared to a current audio-based hand guiding system and delivers overall a more intuitive and relatable fine-point guiding experience.

CCS CONCEPTS

• **Human-centered computing** → **Accessibility**; **Haptic devices**; *Human computer interaction (HCI)*.

KEYWORDS

haptic interface, guidance, localization, tracking, visually challenged people

ACM Reference Format:

Felix Huppert, Gerold Hoelzl, and Matthias Kranz. 2021. GuideCopter - A Precise Drone-Based Haptic Guidance Interface for Blind or Visually Impaired People. In *CHI Conference on Human Factors in Computing Systems (CHI '21)*, May 8–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 14 pages. <https://doi.org/10.1145/3411764.3445676>

1 INTRODUCTION

A recent survey [9] estimates the number of blind people worldwide to be about 36 million. A far greater number of about 216.6 million individuals must cope with moderate to severe visual impairments. Reduced perception capabilities of blind or visually impaired (BVI) people make instrumental activities of daily living (IADLs) [11] (e.g., shopping or working) harder and can take a toll on the mental health

of an individual by causing depression or anxiety [27]. Despite these challenges, visually impaired people try to lead a normal, independent lifestyle [4, 27]. The ordinary task of searching and locating smaller objects in unknown, dynamic surroundings still represents a significant, time-consuming, and sometimes frustrating challenge [29]. The affected group of people often has to rely on either laborious space exploration by touch sensation or third-parties for support [4, 34], which can distance them from their desire for an independent way of living. Technological advancements around human-computer interaction (HCI), computational vision and sensor information processing capabilities provide innovative possibilities to support visually impaired humans in overcoming these daily challenges.

The current state of the art technological aid solutions promise to support the impaired users throughout their mostly independent lifestyle [45].

While audio and haptic feedback depict the center of interest in HCI research for BVI [29], the field has spread into utilizing innovative concepts like augmented reality (AR) or miniature drones to present further haptic and visual enhancements for impaired people [1, 4, 19, 26]. Haptic and tactile interfaces augment the users explore by touch approach [16, 17, 29] by generating new haptic cue sets to encode further localization information [30, 35]. For small-scale, fine-point search and localization interactions (e.g., grabbing an object or tabletop interaction), which amounts to a significant part of daily activities, proposals for technology assisted guidance aids exist (e.g., FingerReader2.0 [8], ShopTalk [34], SmartGloves [30], GIST [25] or Wilson et al. [46]). Many of the existing interface concepts represent highly complex and specific solutions [25, 34, 46], restricting their general purpose applicability. For those proposed systems, the utilized human-computer interfaces are not always ideal, as the provided feedback cues often significantly overlay other sensory channels (e.g., hearing or touch) or require extensive learning of specific feedback patterns [33]. Due to the requirement to translate audio and haptic cues into movements, a high cognitive load is often put on the user while using such an aid. Today, many visually impaired people still heavily rely on their 'analogue' tools and explore by touch capabilities (e.g., white cane, guide dog and touch sense exploration) [44]. There is a clear need for an intuitive, dynamic, general purpose aid device, which augments and supports the BVI user in their IADLs while not impeding their senses and movements in a significant way. In the area of BVI aid devices, multi-modal feedback consisting mostly of a combination of haptic and audio feedback has been proven to be successful [16, 22]. As shown in prior studies [4, 6], drone interfaces offer more direct, thus easier to translate and interpret

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from [permissions@acm.org](https://permissions.acm.org).

CHI '21, May 8–13, 2021, Yokohama, Japan

© 2021 Association for Computing Machinery.

ACM ISBN 978-1-4503-8096-6/21/05...\$15.00

<https://doi.org/10.1145/3411764.3445676>

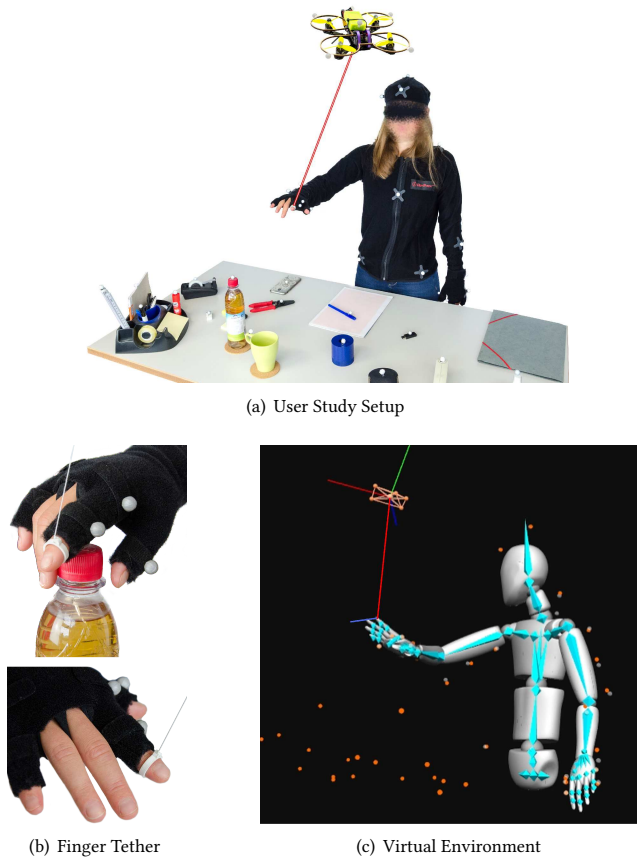


Figure 1: Figure 1(a) shows an anonymized participant in the comparative user study with the HDI guidance system. The human-drone tether connection string is highlighted as a red line. Illustration 1(b) displays details of the used finger tether ring in combination with the OptiTrack marker glove for tracking the users' hand. Screenshot 1(c) showcases the virtual environment with the tracked study participant, drone and study task objects.

guidance feedback without exposing the user to high cognitive load scenarios.

A recent research interest in HCI has been tangible guidance interfaces implemented by using quadcopter drones [6, 19, 26]. Already widely used in virtual reality (VR) research for perceivable force feedback objects and control inputs (e.g., Flyables, Thor's Hammer, Valkyrie or VRHapticDrones) [1, 19, 32], new, more advanced application cases for haptic human-drone interaction (HDI) started to emerge.

Until now, HDI research in the context of BVI aid systems focused mainly on coarse locomotion navigation (e.g., guidance while walking through a corridor). The DroneNavigator research project proposed miniature drones as a digital guide dog for coarse navigation tasks [3, 4, 23]. The unique emitted sound profile from drones provided navigation cues [3, 4], or a physical connection to

the drone exerted haptic feedback cues [6]. Miniature drones offer a well-connected mobile sensor platform, used flexibly in many application areas, independent or dependent of the user [6, 19, 26]. In comparison to typical vibration and haptic touch feedback concepts, drones represent a different variation of haptic feedback as they can exert multi-dimensional pulling or pushing forces on the human body. Our research focuses on the application case of a human-drone tether connection for providing accurate, tactile feedback for close-range, high-resolution search and guiding tasks (c.f. Figure 1). In comparison to the prior drone-based locomotion navigation concepts (e.g., assist while walking) in BVI research, we desire to use a drone for high precision guiding of the users' hand (e.g., help finding a cup in a kitchen). We define the research question for this paper as follows: *Can a drone serve as a fast, accurate and usable aid for augmenting precise guiding tasks within one's arm's reach for visually impaired people?*

Figure 2 illustrates our overall research approach. A focus group of ten blind people complemented our research during the development process and the prototype evaluation user study. We chose to directly work with target group users in the early research stages due to their unique touch and force perception capabilities that we want to leverage in our HDI [37]. In a first meeting, we presented the very abstract concept of utilizing a HDI for fine-point guiding tasks and related work projects to the focus group. With the moderation of two senior researchers, the group discussed our HDI approach and related work contributions to develop an initial concept idea. In a pilot study, the BVI participants evaluated the directional force perception thresholds for each of the defined drone-human tether connection points (cf. Figure 3(a)). Based on the measured perception threshold data and personal insights from the hands-on experience, the focus group later proposed cue patterns for the drone guiding interface. With the concept ideas from the focus group, we implemented a proof-of-concept prototype of the drone guiding system. In the end, ten separate BVI individuals, not familiar with the development process, evaluated the prototype in a controlled lab study and compared it with current audio augmented localization approach, commonly used in related BVI tabletop interaction research.

With the pronounced research of visually impairment aids for general navigation tasks (e.g., walking through a building or city), our research focuses on guiding interfaces in the haptic interaction space (e.g., finding a cup or pencil on a desk, c.f. Figure 1). Our haptic drone interface addresses the common HCI issues of other 'smart' systems by offering an intuitive to use guidance solution for BVI people without significantly restricting the natural sensing capabilities of the user. From our user study, we have shown greater usability and lower cognitive processing requirements when we compare our drone-based approach with the audio guiding approach. In the presented research, we focus on the guiding of the user in an unknown environment. While grasping might also be part of the overall interaction, we did not research this part of the interaction in full detail. Our contribution shows the viable interaction application case of a tethered drones for guiding blind people in precise search and positioning tasks. Our prototype aims at improving the ability of blind people to be more confident and independent in fine-point interaction tasks.

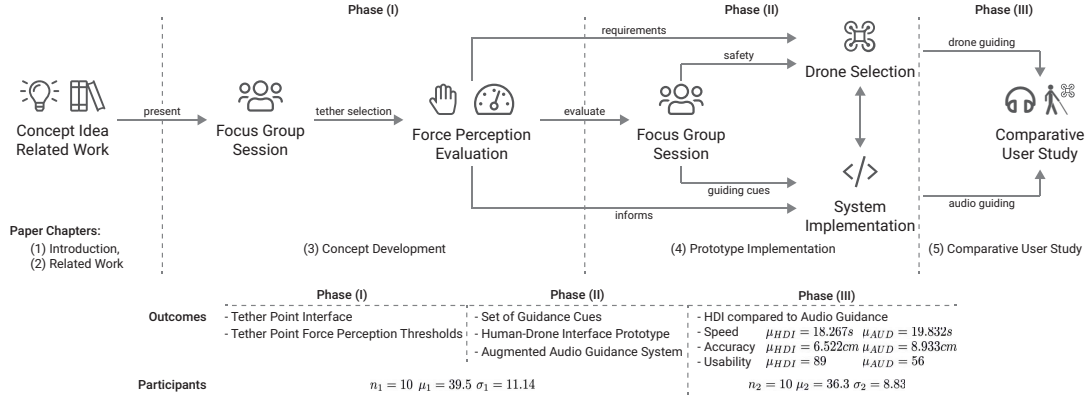


Figure 2: Overview of our research scheme, related paper chapters, summarized outcomes and two participating BVI groups. The sequence illustrates the integration of the BVI focus group throughout the individual research parts, which conclude in the comparative evaluation user study. Phases one and two were conducted in multiple sessions in a span of a couple days. The presented outcome indicates that the proposed human-drone interface is faster, more accurate and has better usability compared to the audio guidance approach.

2 RELATED WORK

Research of technical navigation aids for BVI people has an extensive history, from first technology-assisted personal guidance systems in the early 1990s [38] to the advanced wearable and solutions of today [18]. Traditionally, the impaired utilize 'analogue' aids, such as guide dogs, white canes, or they rely on support from other people for general locomotion navigation [6]. For fine-point search and localization of objects and tabletop interaction, BVI individuals apply the explore by touch approach to 'see' their surroundings through their skilled touch perception capabilities [17, 29]. In the last decade 'smart things', ubiquitous computing and wearables have primarily dominated related research efforts for both locomotion navigation and fine-point localization tasks [21]. With the exploration of human-drone interaction (HDI) for HCI, researchers adapted quadcopter for BVI aid research [4, 6]. The benefit of using drones is the dynamic haptic force feedback and possibility of both input and output modality in various HCI applications [1, 19].

Freundschuh et al. [15] categorized the general human interaction tasks, which were extended by Nicholson et al. [34], into the three spatial interaction spaces: (I) locomotor, (II) search and (III) haptic. The proposed standard spatial dichotomy classifies these three spaces according to the space of interaction [34]. Longer distance navigation and locomotion (i.e., walking through a building) defines the locomotor space [15]. In the search space, only a limited amount of locomotion occurs. The search interaction focuses on a specific area of interest for navigation (i.e., locating an object on a shelving wall) [34]. The haptic space defines all user actions occurring within one's arm's reach, without any user locomotion [15]. A large focus area is the locomotion navigation of impaired people using technological aids with multi-modal interface solutions [4, 31, 48]. With growth of touch interfaces over the last decade, tabletop [25, 35, 46] and touch screen interaction [16, 17, 22] have gained attention and play a vital role in current BVI related research [29]. A common way to make visible cues better perceivable for BVI humans is to utilize other distinct sensory inputs of the

human body (e.g., hearing, touch or haptics). From a technological standpoint, the modality translation includes sensors inputs (e.g., ultrasonic sensors or real-time vision and tracking processing) and some means of output interface, which communicates the feedback to the impaired user [8].

For guidance and localization in the context of touch and tabletop interaction for BVI users the related work focuses on the usage of a wide range of audio interfaces. With headphones or speakers in combination with a smart device, the visually impaired user can gain additional information, essential for the current localization task at hand (e.g., current position, surrounding items or obstacles) [34]. Audio guidance aids integrate a large granularity of guidance possibilities, from step by step route guidance [47], to locating small-scale items [39] with three-dimensional augmented audio setups [13, 14]. FingerReader2.0 resembles an abstract audio interface for BVI users, which provides auditory feedback based on an image from a finger mounted camera. By pointing at a sign or product box, the impaired users get all the text read out to them, thus it offers some directional feedback for locating certain objects [8]. In the NAVIG project [24], researchers deployed a three-dimensional sound rendering system. Katz et al. [24] used this auditory feedback system in combination with advanced camera object recognition to both inform and navigate the study participants. Both coarse locomotion navigation (e.g., moving through a building) and fine-point localization tasks (e.g., grasping an object on a desk) were demonstrated with the NAVIG system. Wilson et al. [46] proposed a similar localization tool for young visually impaired people. Rather than simulating three-dimensional sound, Wilson et al. used speakers mounted to the individual objects as localization cues. Both research projects [24, 46] indicate overall accurate spatial interaction when using the developed spatial audio cues. In addition to simple spatial audio tones, spoken and read out audio cues and directions are used for search and localization applications as well. Kane et al. [22] presented in their research different audio interface options to enable BVI users to interact with touch screens more

easily. Some of the presented approaches (e.g. guided directions or touch feedback) have been applied to augment localization and search tasks outside the touch screen application domain [6, 40]. The related work shows a vast array of auditory feedback interfaces for the application of localization and search tasks.

Compared to audio augmented interfaces, haptic feedback offers the benefit of not blocking the hearing capabilities of the user [6]. The research surrounding BVI aid support system applies the concept of tactile-visual sensory substitution [20]. In the Deep Learning-Based Wearable Vision-System (DLWV2) project [30], researchers used a wrist-mounted camera to identify objects and provided directional cues employing vibrotactile feedback through a wrist-mounted wearable. In their user study, the DLWV2 was faster and more accurate compared to searching without any assistance system using exploration by touch. The DLWV2 system as a whole offers a solution for search and haptic space localization tasks, but the feedback provided is limited [30]. FingerSight [37] consists of a finger-worn camera and four vibrotactile elements inside a ring for fine-point guiding of BVI. While the FingerSight study showed accurate and precise movements guidance were possible, factors like overall user experience or intuitiveness were neglected and not validated with real BVI people [37]. Vibration motors mounted inside a vest, wrist wearable or glove can also serve as output modality for unique directional patterns [20, 29, 30]. Based on the perceived vibrotactile feedback, the user can adapt and adjust their walking route [30]. While vibrotactile feedback might not be the most pleasant tangible user interface (TUI) for communicating directional cues, it promises clear, identifiable patterns, which the user can learn and adapt to [10].

A recent extension of the haptic interface opportunities are miniature drones. The concept of grabbing or touching a drone, representing an object which can exert a force on the user, is used in VR and AR applications [1, 19, 26]. Knierim et al. packaged a drone inside a graspable cube shape to provide a haptic force interaction interface, enabling them to drag the hand of the user to specific locations within the virtual world. The concept of small-scale drones for the augmentation of position guidance was transferred to other research fields. Noise, emitted from a drone, can provide location cues to BVI people [3, 4]. In a comparative user study, the drone audio interface provided more accurate and reliable spatial feedback compared to an audio augmented aid solutions [4]. Soto et al. also developed a tangible drone navigation system, which suggests the usage of a leashed drone tethered to the user [6]. Through the attached handle, the drone can provide directional haptic feedback by 'dragging' and guiding the BVI user along a chosen route. The accuracy of the results was comparable to their auditory drone interface, but the leashed interface proved to be more reliable in noisy environments [6]. Researches see further application areas for refined search and localization interaction as feasible, due to the flexible, dynamic and evolving nature of human-drone interfaces. Another key research topic regarding the usage of drone interfaces is the safety and social acceptance of such interfaces. While many VR/AR and some BVI related projects use drones as interface objects, the insight into safety concerns and impact in the interaction with humans is rather less well documented [6, 36]. The interaction safety plays a vital role in the social acceptance and success of

future application of drones in the HCI field, as it is currently a limitation in many concept prototypes.

Force perception plays a vital role in human-robot, robot-human interaction and soft-robotics, especially for blind or visually impaired people. BVI people heavily rely on the touch sensory channel as means of perceiving the environment [27]. To make a force or touch stimuli perceivable for a human, an exerted sensation has to be larger than an absolute perception threshold [43]. The human force perception is highly dependent on many application factors (e.g., perception point and area size, means of force transmission or individual sensitivity). The human fingers are among the most sensitive to high-resolution touch and force stimuli perception and differentiation [28]. While a large amount of research has focused on the extended touch and push perception capabilities of a human, limited research in the field of pull force perception of BVI is available [12]. In tangible human-drone interaction when drones serve as haptic in- and output modality, force perception is the key interaction point [6, 26]. While large commercial drones can generate significant forces, smaller-scale drones used in VR and HCI applications are so far limited in their force generation capabilities [1]. Drones most often serve as ungrounded, haptic interfaces, where the drone is interfaced by touch or holding, thus stimulating relative force changes on the user's body. In some applications, drones generate pull forces for a user perceivable interface [6]. Scientific research on the human perception capabilities and threshold of pull force stimuli through HDI is so far limited.

3 CONCEPT DEVELOPMENT

The concept development started with a meeting of the BVI focus group (cf. Figure 2). Ten individuals, aged between 28 and 56 years ($\mu_1 = 39.5$, $\sigma_1 = 11.14$), eight male and two female, all right-handed, from a regional BVI community, complemented by two senior researchers as moderators made up the focus group. All the participants are severely visually impaired since birth and had no light perception capabilities according to the World Health Organization (WHO) level 5 classification mentioned in their medical record. To support their daily routines, the participants relied on a wide range of aid solutions (e.g., white cane, guide dog or support worker). Three participants had limited prior experience with technological aid solutions (e.g., smartphone applications and audio navigation system), but none had experience with HDI or other technology-assisted tangible force interfaces. The goal of the first session was to reflect on existing concept ideas for human-drone guidance interfaces based on prior research and expectations from the BVI group. Further, these BVI focus group participants later took part in the separate pilot study session to evaluate possible human-drone tether connection points by means of determining the absolute force perception thresholds. The results of the focus group and pilot study build the theoretical grounded foundation for our HDI guidance prototype implementation. The realistic comparative study (HDI vs. audio guidance) was conducted with ten separate BVI participants.

3.1 BVI Group Concept Session

After a short mutual introduction of the focus group attendees, the two moderators gave a brief introduction to the general HDI

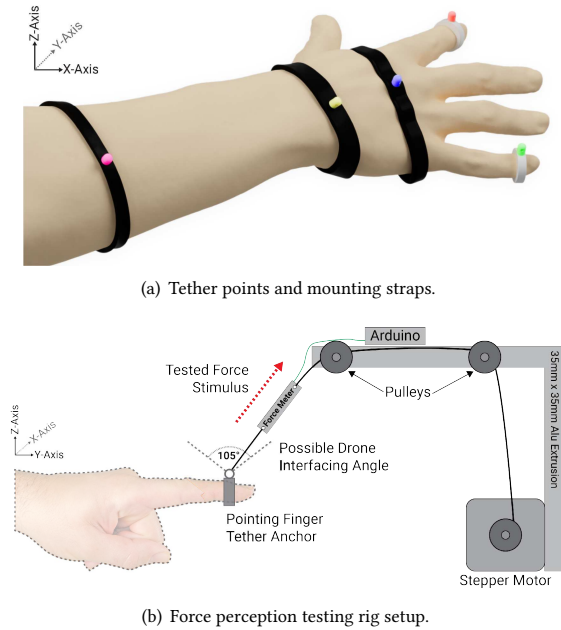


Figure 3: Illustration 3(a) shows the locations and strap mounts of the five drone-human tether connection points. Figure 3(b) illustrates the controlled setup used in the force perception evaluation study to accurately generate directional haptic stimuli.

concept context and related work projects. On an abstract level, a human-drone tether connection exerts directional pulling forces to generate a stimulus interface for fine-point directional guidance. Existing concepts and implementations of HDI in VR environments and BVI aid from the related work projects were outlined to provide a current knowledge base. The moderators provided the group with the abstract idea, as no technical implementation background or detailed concept restrictions were deliberately prepared in advance. The presentation of the research idea was intentionally abstract to keep the participants open-minded for their own ideas. The focus group along with the moderators initiated the interface concept development steps towards the implementation outlined in this paper. The goal of the first moderated group session was to discuss the concept ideas regarding (I) the general concept of how a drone could assist BVI individuals in fine-point guiding tasks and (II) outline requirements for connecting the drone interface to the human body. The two moderators recorded and transcribed the discussions.

The initial response of the focus group participants to the presented HDI idea was consistently positive. Many envisioned a guide dog-like digital companion, a flying guide with an advanced set of guidance features that can 'take them by the hand' in an unknown environment. An important point, brought up by one session participant, was the unrestricted freedom of the hands and fingers for fine-point tasks. A drone guidance interface should not significantly hinder or obstruct the distinct touch sensation capabilities of an individual. This aspect of not blocking the touch sensation was also

pointed out by Soto et al. [5]. During the usage of the envisioned system, hands and fingers should be free to grasp and touch objects without the drone obstructing the user in any way. Existing VR/AR [19, 26] or drone BVI aid [6] interface concepts usually require the user to touch or grab onto a controller or wear gloves [30], thus obstructing ones' touch capabilities. Therefore, these interfaces are less suitable for the fine-point guidance application case. The focus group established and agreed on the concept idea of a tethered flying guide companion and progressed to discussing the suggested human-drone interface possibilities.

The next question at hand was, how should the drone be tethered to the human users to navigate and guide them in an accurate manner. An early concern in the focus group discussion was the safety of the BVI users. An autonomous drone in proximity of the limbs, upper body, and head of users unable to spot the drone, made the impaired participants feel uncomfortable. The moderators noted the participants' general concerns of the drone getting tangled in clothing or hair, thus crashing, or hitting the user. The focus group discussed possible solutions to increase overall drone safety. Besides using only smaller-scale, lightweight drones with propeller guards, the drone connects to the BVI user with a fixed guide rope tether (cf. DroneNavigator [6]). The tether would allow a constant, direct connection while flying the drone further away. Regarding the HDI attachment points, the moderators suggested utilizing arm or wrist tethers based on the prior finding that hand and fingers should be free for interaction. A common conception formed that haptic guidance feedback provided at the point of interaction contributes to a more intuitive interaction with the HDI [28]. The focus group suggested a tether connection at the dominate arm or hand. For a fine-point task, a body part with high sensitivity capabilities to differentiate minor changes in the drone force and direction feedback is desirable [28]. From the brainstorming session and discussions with BVI community members, five possible tether points concepts emerged for our empirical evaluation (cf. Chapter 3.2): index and pinky finger, back of hand, wrist and forearm. Figure 3(a) illustrates the exact positioning and tether mounting methods. The outlined ideas and concepts of human-drone tether connections were tested and validated in the force perception experiments later.

3.2 Force Perception Evaluation

In the pilot study we determined the perceivable directional force thresholds for the suggested drone guidance tether points. Our proposed HDI relies on directional pull forces as guidance cues rather than on commonly researched touch or pressure-based interfaces. The experiment also allowed the focus group participants to experience their suggested tether point interfaces hands-on. Besides the gained quantitative data in terms of force threshold levels, we also collected qualitative interview feedback. For the within-subject study design, the independent variables were (i) the five different tether options, (ii) the directional vector of force and (iii) the applied force level. The dependent variables were (i) the indicated directional force and (ii) the qualitative rating of the tether point attachments. The collected data is of great importance in the design and development of the actual HDI guidance prototype and future HDI systems.

For reference, we use the orientation notation from a top-down view onto the hand; X-axis: left-right, Y-axis: forward-backwards

and Z-axis: upward-downward (cf. label in Figure 3(a)). In order to measure the absolute perceivable threshold of directional force, a controlled setup (cf. Figure 3(b)) to accurately simulate and measure the force stimulus was build. We used a closed feedback loop system based around a pulley-driven setup with a load cell in line with the tether cable. A powerful stepper motor controlled by an Arduino microcontroller generated the pulling force stimulus. As further reference, the study was carried out in a twelve camera OptiTrack environment, which tracked the position and angular relation of the participants' upper-body, finger, and the force generation setup. All five tested tether point mounts consisted of flexible 3D printed straps (e.g. adjustable band or finger ring) with a fixed stimulus anchor point (c.f. Figure 3(a)). The rubber-skin contact prevented the mounts from rotating freely under the exerted force stimuli. Due to the physical limitations of a flying drone, the angular force generation possibilities are limited. From the point of a tether connection point, we defined the usable interaction field to have a cone-shaped interaction space with an opening angle of 105 degrees. We divided the possible directional force output into even increments, resulting in eight possible directions, 45 degrees apart, in addition to straight upwards (cf. Figure 4(a)). Based on related work [30, 37], we valued these nine directional force stimuli to be sufficient to get an understanding of the perception of pulling forces on the limb attached tether points.

The unified position for all participants was standing up straight with their right arm resting at breast height, off to the side of their body. During the study, the force stimulus setup was moved around the user to the fixed, predetermined positions in accordance to Figure 4(a), which were marked in the virtual OptiTrack environment. The arm and hand of the study participant stayed in the same place while the stimulus setup was manually moved by the study supervisors. The OptiTrack system ensured that the angles and positions were comparable throughout the user study testing. Before the study conduction, the study supervisor explained the directional stimulus answer opportunities. In between the study runs, the provided rest breaks counteracted fatigue effects of holding up one's arm for prolonged periods of time. The order of the tested tether points (fingers, back of the hand, wrist, and forearm) and predetermined force angles were chosen at random to reduce occurring sequence effects. We split the presented force stimuli into fixed step-up intervals with $0.025N$ increments and a reset to zero in between. With the Arduino controller, load cell measurements and stepper motor output, the system regulated the force stimulus to achieve a near constant output. After each presented force input, the participant could either indicate a specific direction or unknown label as an answer. We repeated each correctly indicated test condition until five consecutive identical force stimuli resulted in the same output label. This ensured a high confidence in determining the exact perception threshold for the participant. The measurement output of this study were normally distributed force perception levels in Newtons, which represent the minimum force stimulus needed to reliably transfer the outlined directional cues using the different tether point anchors to a BVI individual. For each angle positions normally distributed force stimulus data in Newton was measured. Following the completion of the perception evaluation, the participants were interviewed and rated the tested tether solutions.

The force perception thresholds, as shown in Table 1, were calculated based on measured averaged directional force perception level. Comparing the overall averaged sensitivity of the tested tether points angles, the index ($\mu = 0.374N$, $\sigma = 0.069N$) and pinky finger ($\mu = 0.366N$, $\sigma = 0.078N$) were the most sensitive. Ranked in descending order, wrist, back of hand and forearm followed closely. These observations in the collected data points match the experience and perception of the test participants. During the post-study interview sessions, multiple BVI participants stated that feedback via the forearm or wrist felt more indirect and thus imprecise. In comparison, directional force feedback exerted using a finger tether connection felt more connected and direct. A tightly coupled interface made the haptic feedback easier to perceive and understandable for precise guidance movements. In an overall ranking of likeability and user experience, based on a post-study interview, the finger tethers take the first places, followed by the back of hand, wrist, and forearm in the last place. For the desired HDI guiding application case, the BVI participants stated that forearm, wrist and back of the hand would not be a preferable solution due to the limited sensitivity. All participants favored a drone-finger tether interface for direct haptic feedback in fine-point guiding tasks. The results from the focus group discussion and force perception test are the foundation for the design and implementation of our human-drone guiding approach.

4 PROTOTYPE IMPLEMENTATION

The results from the first focus group session and the force perception study directly inform the prototype development. The second BVI group session focused on the navigation cue concepts. The results of the force perception study provided the baseline specification reference for the construction of an appropriate drone system. The goal was a proof-of-concept human-drone interface prototype for guiding visually impaired persons in fine-point tasks. As part of the comparative user study, we also replicated a commonly used audio guidance approach with spoken auditory directional cues.

4.1 HDI Navigation Considerations

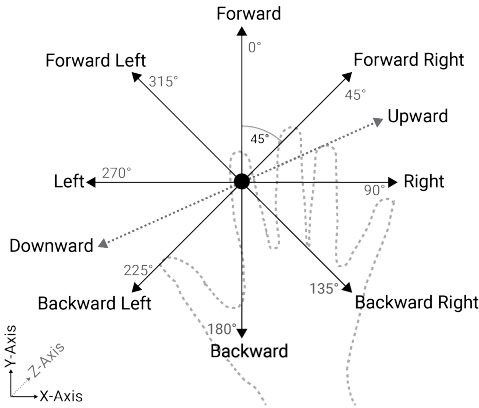
Through the pilot study, the BVI focus group established the index finger to be the HDI tether point of choice. The goal of the second BVI group session was to derive and adapt a set of intuitive to perceive guiding cues for fine-pointing tasks.

Based on the experience from the force perception test, one participant identified the drone always pulling slightly upwards as a potential problem. Due to the physical nature of a drone, it is hard to achieve perfect orthogonal feedback cues. The group agreed that the eight directional cues (cf. Figure 4(a)), as used in the pilot study, were easily distinguishable.

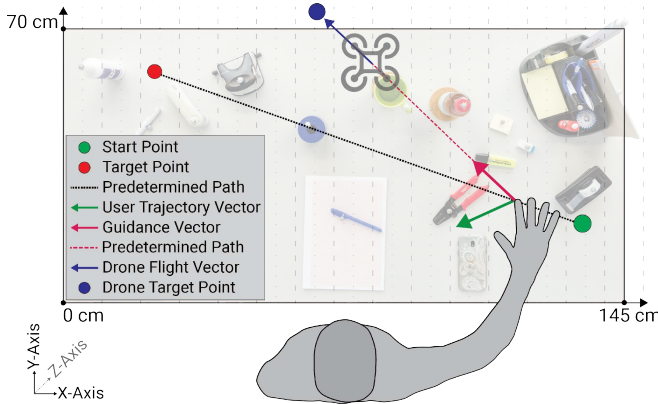
The group of BVI participants extensively discussed the idea to encode the distance to the target object as relative force cue. Related work [28, 43] and the results from the force perception analysis suggest that to differentiate multiple force levels and directions, more significant changes in force magnitude and direction would be needed. Encoding the distance in the applied pulling force would make the overall implementation more complicated and would overload the sensory channel of a novice impaired user. Besides the general issues of the force stimuli differentiation, our

Table 1: The table lists the measured, averaged force levels from the directional perception tests. The zero-degree angle indicates a straightforward Y-axis pull stimulus, the 45° offset measurements follow in a clockwise rotation around the tether point. The last measurement represents the straight upward Z-axis stimulus. In the averaged measurements, the most sensitive perception points (minimal force values) are highlighted.

	0°	45°	90°	135°	180°	225°	270°	315°	Up
Pinky	0.342N	0.370N	0.356N	0.375N	0.402N	0.356N	0.386N	0.381N	0.332N
Pointing	0.339N	0.386N	0.370N	0.391N	0.401N	0.410N	0.355N	0.376N	0.343N
Hand	0.468N	0.502N	0.449N	0.432N	0.486N	0.480N	0.466N	0.488N	0.539N
Wrist	0.647N	0.622N	0.668N	0.666N	0.698N	0.653N	0.601N	0.631N	0.622N
Forearm	0.895N	0.855N	0.820N	0.918N	0.849N	0.878N	0.833N	0.863N	0.797N



(a) HDI direction and augmented audio cues.



(b) HDI guiding algorithm visualization.

Figure 4: In Figure 4(a), the eight possible HDI guiding directions are shown exemplary for the pointing finger drone tether. The wording scheme for the audio augmented system is listed for all eight directions, up- and downwards. The illustration 4(b) visualized our prototype HDI control algorithm, which we use to generate the directional haptic stimulus for the HDI guiding. In the shown case, the drone counter steers the user movement by providing a haptic input in the 'Forward Left' direction, to keep the BVI user on the predetermined guiding path.

focus lies on fine-point guidance, thus we see distance encoding only as secondary feature. The focus group decided towards directional feedback by pulling with a constant force towards the target. Besides the constant force feedback, directional changes in XY orientation still need to be easily perceivable. The drone flight path must, therefore, present a significant, perceivable angular change to guide around sharper bends and turns. To simplify the human-drone guidance approach, the BVI participants suggested to separate the XY- and Z-axis movements. At one point of time, the HDI user would only receive either directional XY- or Z-axis feedback.

A further concern was the implementation of the Z-axis. While the upward movement is natural to encode, providing a downward moving cue with a drone is challenging with a tethered drone. The transition of a drone from above the hand to below is not viable due to time delay, possible space restrictions and safety concerns. Two solutions for encoding the Z-axis information were discussed within the BVI focus group: (I) applying a constant upward force and releasing this force to indicate a downward movement or (II) a short upward jerk to indicate a downward movement and using upward resistance feedback to stop the lowering of the hand. Concerns arose that the difference between a continuously applied force and an upward movement indication would be too insignificant. Furthermore, a constant force application could impact the general force perception capabilities of the user. The user could get used to the constant stimulus; thus the just noticeable difference (JND) would be too insignificant to detect a change in HDI guidance. After trialing the concept ideas with the force perception test setup, the BVI group deemed solution (II) to be more dependable.

In conclusion, the encoding of HDI guiding and audio cues is shown in Figure 4(a). In the user study, starting from a fixed point, the drone would guide the impaired user along the predetermined path towards the target location. During the guidance, XY- and Z-axis movements would be encoded and executed separately. The XY-axis guiding relies on the eight directions with constant pulling force towards the target. The system encodes Z-axis movement with the drone either pulling straight up or a sharp upward jerk indicating a downward movement while using the regular upward force as stopping cue. To indicate that the desired object has been reached, the drone guidance system lets the user nudge into the object thus creating a physical touch reference point. From there the user could use their touch capabilities to readjust their grip and grasp the object. The drone then hovers above the users' hand

and the tether connection is relaxed to ease-off the haptic force feedback.

4.2 Proof of Concept Implementation

As part of the concept evaluation, we implemented an autonomous drone guidance prototype setup for the BVI study participants to experience the HDI guidance hands-on. Furthermore, we replicated a current, commonly applied audio guidance setup [4, 22] to compare with the drone localization approach.

The implementation of the HDI for guiding BVI people is based around a custom-build small-scale drone. We took inspiration from the system designed by Abtahi et al. [2]. A lightweight, small-scale three-inch, brushless racing quadcopter was built with dimensions of 20cm by 20cm wide and 5cm high. The system is running on a 4s LiPo battery for maximum lifting performance. With an peak force generation of about 1.3N ($\mu = 1.258N, \sigma = 0.092N$) upwards and 0.85N ($\mu = 0.856N, \sigma = 0.078N$) at a 50-degree tether angle, our drone can generate haptic force feedback that is always above the previously measured force perception thresholds (cf. Table 1) of the BVI participants. A PX4 compatible Airbot Omnibus F4 [42] flight controller represents the brain of the drone and communicates using a FrSky telemetry radio with the base station computer. Similar to Abtahi et al. [2], we designed and installed a protection in form of a simple propeller guard around the drone for user safety (cf. Figure 1(a)). We added retroreflective markers on the frame to track the position using the OptiTrack environment. In order to measure the performance and force exerted on the tether connection, we added a small load cell with an attachment point for the tether rope. A thin, 50cm long tether string connected the drone to the finger ring tether.

In early stages of development, we used ArduPilot [41] in combination with our OptiTrack tracking and Unity control environment but changed to a similar setup as used by Abtahi et al. [2]. The primary system differences are the usage of our 12 camera OptiTrack system for location tracking and the addition of a load cell to monitor the tether pull force. A spatial overview of the study setup with the OptiTrack camera system is shown in Figure 5. The PID controller inside the drone handles general stabilization. The bi-directional radiofrequency (RF) communication system exchanges the location control commands and force measurements. The OptiTrack system tracks the drone, tether cable, participant hand and upper body and feeds the gathered data into the Unity environment.

Inside our virtual Unity environment, we implemented a wizard of oz control algorithm in the sense that the target location, starting point and 3D guiding paths are preplanned and setup by hand. By default, our algorithm focuses on the XY-axis movements and only switches to the Z-axis mode if a Z deviation of more than 15cm is measured or an object collision would occur. A user guidance vector is calculated based on the tether cable angle, user hand position and measured haptic force. Applied low-pass filters and window filters helped to smooth out high frequency jumps in the tracking and homogenize the haptic force output. The goal of our algorithm is to keep the guidance vector aligned with the preplanned path. By providing corresponding directional haptic feedback through the drone interface, we steer the user towards the target. The Euclidean distance expresses the deviation between the preplanned path and

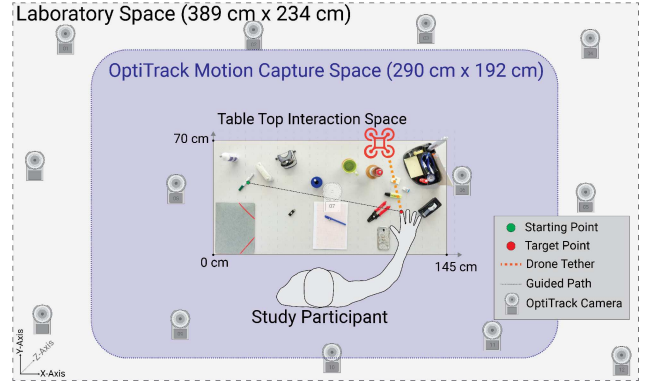


Figure 5: This overview illustrates a top-down overview of the tabletop experiment study setup. In the figure the twelve ceiling mounted OptiTrack camera positions as well as the covered interaction space (Lab space 389cm x 234cm, OptiTrack space 290cm x 192cm) are visualized to give a better understanding about the general study scene.

actual finger position. This deviation is factored into the guiding vector and actively counter steered by the drone.

The drone aims to generate a pulling force of 0.5N by trying to fly to a set target waypoint, which is outside of the tether restricted operation space of the drone. The greater the distance to the set waypoint target, the more power the drone will apply to increase the speed to reach the given point. In Figure 4(b), the fundamental working principle of the algorithm is illustrated. In the ensemble with the tether force measurement sensor feedback and a rudimentary control loop, our drone system can produce a near constant directional force stimulus. The system pursues a constant force output significantly above the set minimal perception threshold level of around 0.5N. The directional drone movement is restricted to the eight predetermined directions shown in Figure 4(a). Furthermore, the angle between the drone and tether point is kept at around 50 degrees for directional cues and only deviates if an object or person would interfere or in case of upward guidance. The movement speed of the drone was limited to $1.0 \frac{m}{s}$ and the maximum acceleration was $0.5 \frac{m}{s^2}$. During 20 test runs, we measured an average position error of around 7cm ($\mu = 7.01cm, \sigma = 2.58cm$) and rotational error of about 3.5° ($\mu = 3.55^\circ, \sigma = 1.86^\circ$). The force generation error for a set target stimulus of 0.5N was $\mu = 0.086N, \sigma = 0.032N$. Overall, the capabilities of our system matched to the comparable setup from Abtahi et al. [2].

4.3 Audio Augmented System

While the typical explore by touch approach used by many BVI people would be suitable to solve the study general task of locating an object on a tabletop setup, the resulting time measurements would be skewed. Due to the learning effect of memorizing the items, the blind user would learn the positions and get faster with time. Even if all objects would be randomized after each task completion, there could be a skew in the collected data result. Furthermore, tactile search localization does not compare well to our conceived HDI application case. For the comparative user study, we desired an audio interface which relies on a similar fundamental localization

guidance principle and focuses on a similar application case as our drone interface. The user should be fully guided by the audio feedback and there should be no need for manual haptic exploration of the unknown environment space. Such directional audio guidance systems have successfully been used in related tabletop [22] and drone navigation [6] comparative studies. Our audio feedback system also closely relates to the *Guided Directions* described by Kane et al. [22]. The proposed system used distance and angular measures to select the corresponding audio cue. The continuous, spoken feedback cues compared to tone direction mapping results in an easier to interpret and learn instruction set [22]. A similar audio guidance approach was adapted by Soto et al. [6] for locomotion navigation. As reference for the selection of the right audio cues, the same OptiTrack environment (cf. Figure 5) is used to track all objects and the user in the environment. Instead of calculating a force-dependent waypoint for a drone, we translate the guidance vector into short, prerecorded voice commands. The voice commands match the ten possible directional drone interactions (8 directions, up- and downward) (cf. Figure 4(a)) and an additional stop command. The audio system gives the XY-axis and Z-axis movement instructions independently. The Unity control PC transmits the commands wirelessly via Bluetooth headphones. In addition to letting the user nudge into target object, the audio system plays an audio cue once the user touches the desired item.

5 COMPARATIVE USER STUDY

For the evaluation of the achievable guidance accuracy and general viability of our proposed human-drone interface, we conducted a comparative user study. We compared our drone guidance system to an audio augmented localization system.

5.1 Study Conduction

We conducted a comparative study with ten BVI people that were not familiar with the prior work of the focus group. The individuals, six male and four female, were between 24 and 44 years old ($\mu_2 = 36.3$, $\sigma_2 = 8.83$), right-handed and severely visually impaired since birth (no light perception, WHO level 5). All participants got a short introduction about both our HDI guiding system as well as the audio augmented aid. For the task at hand, twenty everyday office objects (e.g., cup, bottle, pen, small notebook, etc.) were placed in defined locations on a standard 145 cm by 70 cm table (c.f. Figure 1(b)). We elevated some objects by up to 30 cm, representing objects sitting on top of other objects (e.g., stapler on top of the printer). All objects were easily reachable within the haptic space of the participant and captured with one marker in the OptiTrack system. For each study task, the drone or audio system respectively guided the BVI participant to a selected object. The predetermined paths for the guiding methods were setup to let the participant bump into the target object thus establishing a physical touch contact point. Based on the indicated reference point, the participants could easily adjust their grip and finally grab the targeted object.

To prevent participants from rushing through the tasks, we explicitly mentioned that they should let the system guide them through the task and not try to explore on their own. The independent variables were (I) the targeted object locations and (II) the two used navigation methods. As depended variables we collected (i)

Table 2: Mean comparison of user study task completion time, root means squared (RMS) error and scaled usability score (SUS) of the drone and audio augmented guidance systems. In a t-test with a significance level of $\alpha = 0.05$, the HDI offers a faster, more accurate and usable guiding experience compared to the audio augmented system.

Method	Task Time		RMS Error		SUS Score	
	Mean	P-Value	Mean	P-Value	Mean Score	P-Value
Drone	18.267s	0.0120	6.522cm	0.0015	89	0.0001
Audio	19.832s		8.933cm		56	

the total task completion time, (ii) the taken guide path, (iii) errors made by participants and (iv) the perceived usability experience. The randomly sequenced target locations were identical for the study runs with both guiding approaches to obtain a comparable data set. Due to the within-subject study design, we rearranged the objects in an fixed alternative setting for the second guiding method (e.g., identical location, different object).

At the beginning of the user study, the ten BVI participants got a short introduction about the tasks at hand. We conducted two exemplary test rounds where the participants could explore the operating principle of the guiding methods with a short follow-up discussion about potential questions. After the introduction to the systems, the participants took part in a total of ten runs per guiding method. The order of the two used guiding methods was randomized for each participant to counterbalance any sequence effects. For each of the study conditions only the applicable and required setup (e.g., drone harness or audio system) was worn by the participant.

In between the individual study runs, ten participants could regenerate in short breaks and we swapped the batteries on the drone to ensure comparable performance. The participants stood in the center in front of the desk and the right hand was at the default starting position. The study supervisor manually activated each guiding system. The supervisor would disarm the system and reset the study run once the target object has been grabbed or in case of imminent dangers. The BVI participants completed an adapted SUS questionnaire [7] and took part in a short review interview at the end of the user study. We opted for the short SUS questionnaire and open interview session in the end, as interviews have shown to be more beneficial compared to longer, cumbersome to fill out questionnaires in our past BVI research. The OptiTrack motion tracking system recorded the movement traces, task completion time and the search path.

5.2 Human-Drone Guidance

The ten participants successfully completed all drone guiding sessions without any incident or issues (e.g., drone hitting participant or participant bumping into undesired object). The tracking data shows that users let the system guide themselves and did not try to overrule the drone interface (e.g., moving in opposite direction to drone force stimulus). The participants showed no negative reaction with being nudged into the target object by the drone guiding interface and subsequently grasping the item. Figure 6 illustrates an

example track of the collected HDI path data compared to the pre-planned track. In most of the recorded data points, a slight upward trend throughout each given task is observable. We listed the task completion time statistics and calculated root means squared (RMS) distance deviation from the set guide path in Table 3. We summarized the collected data and compared the two tested methods in table 2. The task completion time for our drone guidance interface was $\mu_{HDI} = 18.267\text{ s}$, $\sigma_{HDI} = 5.188\text{ s}$. The averaged measured RMS deviation for the HDI was $\mu_{HDI} = 6.522\text{ cm}$, $\sigma_{HDI} = 1.286\text{ cm}$. The system usability questionnaire (SUS) evaluation presented an averaged rating of $\mu_{HDI} = 89$ with a standard deviation of $\sigma_{HDI} = 5.48$ points for the drone guiding interface 2. A rating of 89 is considered above average and indicates overall good system usability [7].

In the post-study, semi-structured interview, the study participants positively reflected on their personal experience with the drone guiding system. The haptic navigation cues were described as 'intuitive, unobtrusive and easy to grasp after a short learning phase'. P4 mentioned that the provided haptic feedback resembles to 'being guided by someone else's hand' and described his experience as 'very pleasant'. It was noted that having a tether mounted on the finger made grasping an object securely a bit more challenging. The availability of direct haptic feedback gave the impaired participants the feeling of being fully accurate with their movements. P3 and P7 pointed out that the switch between XY- and Z-axis guiding 'breaks the otherwise seamless experience' of 'been taken by the hand'. Nine of the ten study participants stated feeling 'comfortable' and 'safe' while using the drone guiding system. Participant P5 and P9 were uncertain at first about the drone, 'possibly hitting' them and stated being 'anxious about the noise and draft of the drone'. Throughout the trial and study runs, the participants 'gained trust' in the system driven by the 'positive experience' of completing the first couple tasks. All ten participants enjoyed the experience of being guided by a drone and some could imagine using a similar system in a future refined format.

5.3 Audio Augmented Guidance

The usage of the audio augmented guidance system resulted in faster, but more rough, jerky movements with overall larger deviations from the preplanned track. The reaction of the users to a new audio cue transition leads to these quick changes in movement direction and speed. In total, we recorded thirteen incidents where a participant touched an unwanted object or the desk during navigation movements in contrast to zero incidents with the HDI. The undesired items were not further regarded by the participants due to the missing audio cue which indicates not reaching the target item in these occasions. Figure 6 shows such an exemplary tracking data record from the audio augmented guiding compared to a track with the HDI. Longer range, linear movements were accurate and fast. For the final approach and fine-point guiding, some participants circled around the target object. The task completion time for our audio guidance approach was $\mu_{AUD} = 19.832\text{ s}$, $\sigma_{AUD} = 6.426\text{ s}$. The averaged measured RMS deviation for the audio system was $\mu_{AUD} = 8.933\text{ cm}$, $\sigma_{AUD} = 2.318\text{ cm}$. In the SUS questionnaire, the audio guiding approach rated by the participants on average $\mu_{AUD} = 56$ points with a standard deviation of $\sigma_{AUD} = 7.79$ [7].

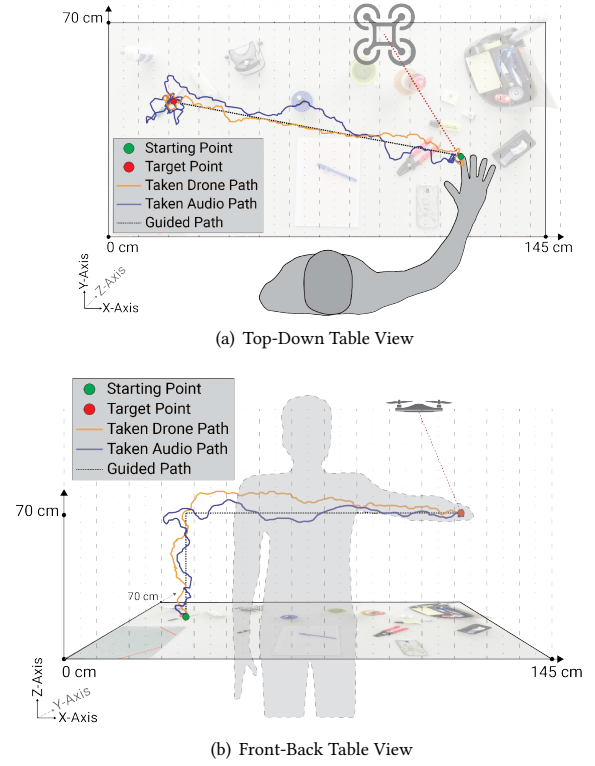


Figure 6: Figures 6(a) and 6(b) illustrate sample raw tracking data of comparable study runs with both the audio augmented (blue path) and HDI (orange path) guiding system. We picked the two shown data recordings since both represent the general observed guiding patterns of the used guiding approaches very well. In the HDI track, the slight, constant upward drift but overall, more accurate following of the preplanned guidance track is observable. The track from the audio augmented guiding highlights the larger deviations around the target area.

The below-average usability rating was reflected in the interview sessions statements. The participants described the audio augmented guiding system as 'novel, but less intuitive' to use. P1 describes 'poking in thin air with additional audio cues' as 'very frustrating' and would instead rely on 'exploring the space using touch sensation' alone. Many of the impaired participants felt insecure and 'left alone' in mid-air without having 'something to touch or guide' by for reference. P8 and P10 suggested being 'less accurate' based on their experience of bumping into objects more often. Both participants P3 and P10 mentioned that the audio cues 'took away their attention' to the surrounding environment, as the focus shifted to concentrate on the audio cues and their movements. Seven participants had minor issues 'translating the audio cues' into actual directional movements. P5 summarized that he had no idea if he 'was following the path or going around in circles correcting

my errors'. In conclusion, eight of the ten participants would prefer touch only exploration over the audio system. The BVI users could imagine audio navigation systems working for locomotion tasks as shown in related work but deemed this approach as less suitable for fine-point guiding tasks. Some study participants pointed out the usage of the audio interface to augment touch exploration as shown in related work but criticized that the localization speed in unknown environments would be even worse compared to the drone interface.

6 DISCUSSION

In our realistic application user study, we compared our prototype human-drone interface against an existing audio augmented guiding solution from literature regarding speed, accuracy, overall usability, and likeability. In terms of overall task completion time, the human-drone interface was significantly faster $\mu_{HDI} = 18.267\text{ s}$, $\mu_{AUD} = 19.832\text{ s}$, $p = 0.0120$, $\alpha = 0.05$. The observed movement speed while using the audio augmentation system was higher compared to the HDI. From our observation, the faster and jerkier movements resulted in less precise movements thus the participants took longer to complete the localization tasks. Comparing the task completion times on an individual task basis, the time advantage of the drone guidance interface becomes more pronounced in longer and complex guiding paths (e.g., T1 vs. T9). For the audio guidance system, we identified a correlation between the task completion time, RMS deviation and the overall complexity of the individual task. Using the audio guidance, a higher complexity and longer task time leads to more deviation from the described route. As for the drone interface, we did not identify a clear correlation between our measures and the given tasks. The complexity of the task does not seem to impact the overall accuracy of the drone guidance. The individual movements showed a smaller RMS deviation error with the HDI ($\mu_{HDI} = 6.522\text{ cm}$, $\sigma_{HDI} = 1.286\text{ cm}$), with the audio guide system ($\mu_{AUD} = 8.933\text{ cm}$, $\sigma_{AUD} = 2.318\text{ cm}$) presenting a higher variance in fine-point guiding situations. The interaction with the drone was more gentle, based on the average movement speed ($\mu = 8.933\text{ cm}$, $\sigma = 2.318\text{ cm}$) and maximum acceleration ($\mu = 8.933\text{ cm}$, $\sigma = 2.318\text{ cm}$) observed. Comparing the individual participants with each other, there are obvious differences in the task completion time and achieved accuracy. Based on our observations and the data set, most of these discrepancies could be traced back to the different behavior of the individual participants. Some participants took a careful approach while completing the tasks, thus taking longer with both compared systems. It was also observable that many participants gained trust in the systems over time, thus their movement speed and task completion time increased, but in most cases only by an insignificant amount. The tracking data (cf. Figure 6(a)) reflects the statements of the BVI participants. It was harder to accurately locate an object using the audio setup in the unknown environment. The main benefit of the drone interface for visually impaired people is significantly enhanced usability and user experience. The t-test evaluation of the SUS questionnaires indicates a clear preference for the prototype HDI in terms of overall user experience $\mu_{HDI} = 89$, $\mu_{AUD} = 56$, $p = 0.0001$, $\alpha = 0.05$. This result is backed by the feedback from the individual interviews sessions, where the study participants favored the drone-based guiding

approach. The summarized data results are listed in Table 2. These results indicate that with the help of the human-drone interface the average task completion time is shorter using the drone and offers a more accurate and better usable fine-point guiding experience compared to the audio augmented localization guidance approach.

Based on the results, we accept our alternative hypotheses.

In the context of locating objects fast and accurately in unknown surroundings, the audio guide presented a novel, feasible approach for tabletop interaction, but the impaired users often struggled and felt lost at some point of the guiding process. In contrast, being actively pulled by the haptic drone interface 'felt more like being taken by the hand of a person'. When asked to compare both tested guiding approaches, all ten participants favored the drone interface over the audio guide. P2 specifically pointed out that his 'movement had a direct impact on the drone behavior and the force interaction'. Two participants saw the audio augmentation as additional high-level input that could easily be ignored, and the task completed by touch sensation alone although being significantly slower. Providing only directional auditory cues, opens space for interpretation. A misinterpretation can lead to the user misjudging a movement cue, thus hitting other close by objects resulting in an overall unpleasant, frustrating experience in the fast exploration of an unknown environment.

The HDI system demonstrated clear benefits compared to a commonly used audio guiding system for BVI. The developed HDI approach for blind or visually impaired people offers an intuitive aid solution for a wide range of application cases. The intuitive, tangible sensation generated by the drone requires little to no adaptation by the user. Unaccustomed users can grasp the concept behind the drone interface right away without the need for too detailed instructions. The users in the study felt more connected with the proposed drone-based aid device. The closer bond to the drone guiding aid provided a secure and trustful environment for the BVI.

6.1 Current Drone System Limitations

In our research, we aimed at exploring new tactile interface possibilities using a drone to precisely guide blind and visually impaired users in fine-point localization search tasks. In the current state, our physical interface prototype consists of a fixed lab setup, which limits the real-world applicability. A challenge of the human-drone interface operation is the battery run time, which has an impact on the overall real-world applicability. In our study, we swapped the rechargeable battery after three to five minutes of continuous use depended on the current state of charge. Further challenges for the full real-world implementation outside a controlled lab are the real-time data transmission, processing and controlling, as well as the spatial tracking without an external tracking setup. The currently needed drone and user tracking requires a 50.000 Euro fixed external tracking system, which takes three hours to setup for 15 square meters of usable interaction space. All guiding paths are preprogrammed and controlled by an additional operator. The present associated drone companion mounting setup and maintenance is also not feasible for an BVI user and requires assistance from another person. These technical application challenges are valid for a wide range of current HCI research approaches, not only for BVI focused work. Another aspect is the overall safety

Table 3: The table lists both the task completion time in seconds and averaged root-mean-square (RMS) deviation in centimeter from the set guidance path for each of the ten study tasks. The average task completion time of our HDI approach was $\mu_{HDI} = 18.267 s$, $\sigma_{HDI} = 5.188 s$ and $\mu_{AUD} = 19.832 s$, $\sigma_{AUD} = 6.426 s$ with the audio augmented setup. Using the HDI approach the average RMS deviation was $\mu_{HDI} = 6.522 cm$, $\sigma_{HDI} = 1.286 cm$ and $\mu_{AUD} = 8.933 cm$, $\sigma_{AUD} = 2.318 cm$ utilizing the audio setup.

	Task	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10
Audio	Time in sec.	10.87	23.36	22.48	18.84	16.21	11.56	28.32	14.44	28.26	23.98
	RMS dev. in cm	8.18	7.33	7.98	11.96	6.25	9.22	5.50	10.73	12.43	9.75
Drone	Time in sec.	11.51	19.18	20.44	20.01	14.20	11.48	25.99	13.98	24.31	21.57
	RMS dev. in cm	7.14	5.46	8.80	4.67	7.55	5.49	6.01	7.74	6.76	5.60

and social acceptance of the generated noise and potential dangers emitted from the drone. This can be viewed in both regards to the user and external bystanders. While our system is intended to be only used in short bursts, thus minimizing the duration of noise emissions, these factories play a key role for the successfully implementation of the technology. Related work [4] deemed the benefits for BVI and potential social and safety problems as well-balanced and comparable to other means of aid methods for BVI [36]. In their assessment of social acceptability of drone guiding systems, Soto et al. [5] showed a great acceptance response of BVI users and the general public when informed about the drone usage. In our explorative user study, the systemic factors noise, wind and imminent danger of a flying drone were a concern at the beginning. The users got used to the drone interface by adapting or suppressing factors like wind and noise since no active countermeasures were used in the study. In the concluding interview, the participants outlined that they felt safe around the drone, knowing it was encased and the approximate whereabouts could be estimated based on the tether rope connection. The opinion of the BVI study participants emphasizes that our drone companion interface presents a viable and safe to operate aid system. Nevertheless, current crucial safety factors (e.g., knocking over items or injury of user or bystander) and their direct impact on the drone usage need to be further researched and explored in future work as they represent a major limitation of the current system. Another current limitation is the 3D printed tether ring connections. These rings on the pointing finger could potentially impact the grasping process negatively. The anchors should be further improved and the effect on the grasping process further studied. All these current limiting factors are subject for future research, where our developed human-drone interface prototype could be integrated into a applicable aid for BVI.

6.2 Envisioned Application Case for Future Research

We assume a BVI is situated in an unfamiliar, highly dynamic environment (e.g., supermarket or new office environment). The aid device user wants to quickly find a specific object (e.g., fruit/vegetable, or certain product) within his surroundings without exploring the entire unknown space by hand, feeling and identifying most other objects. The impaired user is carrying his wrist-mounted personal drone companion. With a simple voice command, the companion is queried to help locate the desired item. In the first stage, the drone companion wearable takes off without the human-drone tether being attached. From there the drone uses its integrated depth cameras and sensors to map and search the surrounding area. The captured

data is sent in real time over an internet of things (IoT) network to a cloud server for artificial intelligence (AI) assessment. The drone returns to the user, which holds up his arm to let the drone land on the wrist mount. Afterwards, the drone provides voice feedback if the object has been located. Next, the user can initiate the guiding process with another voice command. From the wrist mount, the drone locks on the human-drone finger tether and starts providing haptic directional feedback to navigate the user towards the queried item. Depending on the starting position of the companion user, the process might also include locomotion navigation guidance (cf. Soto et al. [6]). For the final, high precision localization task in the haptic space of the user, our validated guidance cue approach is applied. Once the object is grabbed, the user initiates the drone to return to the wearable wrist mount and charges back up for the next assignment. This proposed drone companion assistant system is a highly dynamic aid device with wide application range. It can be used for fast and intuitive localization of objects, general navigation tasks in the locomotive space and serve to augment the normal touch exploration of BVI individuals. We also think that our shown approach and drone guidance methodology could be integrated and used in the future to further augment existing VR/AR experience setups. Our haptic drone interface tethered to the hands of the user could help to create a more immersive VR/AR experience.

We believe that the advances in wearable and smart technology will help to successfully conquer both the technical and social implementation challenges ahead. We see our first prototype as a proof of concept that drone interfaces could make a viable contribution to the haptic human computer interaction through direct tether feedback systems. We envision an affordable, wrist-mounted drone companion that can be easily be deployed in seconds and is tailor made for the needs and capabilities of a BVI individual.

7 CONCLUSION

In this paper, we conceptualized, implemented, and tested a human-drone interface to support visually impaired people in everyday accurate guiding tasks in unfamiliar surroundings. We developed a human-drone interface concept in close collaboration with a focus group of ten blind or visually impaired people. In a pilot study, we evaluated the minimal pulling force thresholds to produce meaningful directional guiding cues for visually impaired persons. In a realistic application scenario, the human-drone interaction was compared with ten BVI participants against a current audio augmented guiding system from literature.

With the collected data and evaluated data (cf. Table 2), we answered the initial research question. The drone interface offers

a significantly more accurate, faster, and more usable guiding interface modality for the exploration of unknown, dynamic environments. Compared to a common audio augmented system, our drone system provides a tangible interaction, improved accuracy, faster interaction speed and better user experience. The perceivable, tangible interfacing experience led to a more connected sensation, which made the study participants feel secure and resulted in a more believable and trustworthy technology interaction. For the integration of ubiquitous computing devices in everyday life scenarios, human-machine interaction is a crucial factor, especially for visually impaired people, who are limited in their visual sensory input. With the drone guiding aid, we provide visually impaired people with a way to localize, guide to and grasp small objects within arm's reach safely and accurately through a reassuring, usable and tangible user interface. We outlined the current limitations of the implemented prototype, but also presented our visions for the possible future application cases in the context of wearable, companion drone aid devices. With our work, we extend the foundation for further expansions of the tethered human-drone interfaces towards complete user guidance and assistance in locomotion, search, and haptic space tasks in both real- and virtual-world applications. The presented approach helps to minimize the daily challenges in the unknown haptic spaces faced by blind or visually impaired people by utilizing current, state-of-the-art technology, thereby contributing an important part towards a solution for a more independent life and more self-reliance of BVI people.

REFERENCES

- [1] Muhammad Abdullah, Minji Kim, Waseem Hassan, Yoshihiro Kuroda, and Seokhee Jeon. 2017. HapticDrone: An Encountered-Type Kinesthetic Haptic Interface with Controllable Force Feedback: Initial Example for 1D Haptic Feedback. In *Adjunct Publication of the 30th Annual ACM Symposium on User Interface Software and Technology* (Québec City, QC, Canada) (UIST '17). ACM, New York, NY, USA, 115–117. <https://doi.org/10.1145/3131785.3131821>
- [2] Parastoo Abtahi, Benoit Landry, Jackie (Junrui) Yang, Marco Pavone, Sean Follmer, and James A. Landay. 2019. Beyond The Force: Using Quadcopters to Appropriately Objects and the Environment for Haptics in Virtual Reality. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland UK) (CHI '19). Association for Computing Machinery, New York, NY, USA, Article 359, 13 pages. <https://doi.org/10.1145/3290605.3300589>
- [3] Majed Al Zayer, Sam Tregillus, Jiwan Bhandari, Dave Feil-Seifer, and Eelke Folmer. 2016. Exploring the Use of a Drone to Guide Blind Runners. In *Proceedings of the 18th International ACM SIGACCESS Conference on Computers and Accessibility* (Reno, Nevada, USA) (ASSETS '16). ACM, New York, NY, USA, 263–264. <https://doi.org/10.1145/2982142.2982204>
- [4] Mauro Avila, Markus Funk, and Niels Henze. 2015. DroneNavigator: Using Drones for Navigating Visually Impaired Persons. In *Proceedings of the 17th International ACM SIGACCESS Conference on Computers & Accessibility* (Lisbon, Portugal) (ASSETS '15). ACM, New York, NY, USA, 327–328. <https://doi.org/10.1145/2700648.2811362>
- [5] Mauro Avila Soto and Markus Funk. 2018. Look, a Guidance Drone! Assessing the Social Acceptability of Companion Drones for Blind Travelers in Public Spaces. In *Proceedings of the 20th International ACM SIGACCESS Conference on Computers and Accessibility* (Galway, Ireland) (ASSETS '18). Association for Computing Machinery, New York, NY, USA, 417–419. <https://doi.org/10.1145/3234695.3241019>
- [6] Mauro Avila Soto, Markus Funk, Matthias Hoppe, Robin Boldt, Katrin Wolf, and Niels Henze. 2017. DroneNavigator: Using Leashed and Free-Floating Quadcopters to Navigate Visually Impaired Travelers. In *Proceedings of the 19th International ACM SIGACCESS Conference on Computers and Accessibility* (Baltimore, Maryland, USA) (ASSETS '17). ACM, New York, NY, USA, 300–304. <https://doi.org/10.1145/3132525.3132556>
- [7] Aaron Bangor, Philip Kortum, and James Miller. 2009. Determining What Individual SUS Scores Mean: Adding an Adjective Rating Scale. *J. Usability Studies* 4, 3 (May 2009), 114–123. <http://dl.acm.org/citation.cfm?id=2835587.2835589>
- [8] Roger Boldu, Alexandru Dancu, Denys JC Matthies, Thisum Buddhika, Shamane Siriwardhana, and Suranga Nanayakkara. 2018. FingerReader2.0: Designing and Evaluating a Wearable Finger-Worn Camera to Assist People with Visual Impairments while Shopping. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 2, 3 (2018), 94. <https://doi.org/10.1145/3264904>
- [9] Rupert RA Bourne, Seth R Flaxman, Tasanee Braithwaite, Maria V Cicinelli, Aditi Das, Jost B Jonas, Jill Keeffe, John H Kempen, Janet Leasher, Hans Limburg, et al. 2017. Magnitude, temporal trends, and projections of the global prevalence of blindness and distance and near vision impairment: a systematic review and meta-analysis. *The Lancet Global Health* 5, 9 (2017), e888–e897. [https://doi.org/10.1016/S2214-109X\(17\)30293-0](https://doi.org/10.1016/S2214-109X(17)30293-0)
- [10] Anke M. Brock, Slim Kammoun, Marc J.-M. Macé, and Christophe Jouffrais. 2014. Using wrist vibrations to guide hand movement and whole body navigation. *i-com* 13 (2014), 19–28. <https://doi.org/10.1515/icom.2014.0026>
- [11] Michael Brooks and Barry C. Stephens. 2014. The Relationship of Age to Level of Performance and Independence Associated with Rehabilitative Interventions Provided to Older Adults Who Are Blind. *International Journal of Social Science Studies* 2, 4 (October 2014), 28–39. <https://ideas.repec.org/a/rfa/journal/v2y2014i4p28-39.html>
- [12] H. Chinbe, T. Yoneyama, T. Watanabe, and M. Nakada. 2016. Haptic threshold for pulling force feedback on surgeon's fingertip in medical robotic systems. In *IECON 2016 - 42nd Annual Conference of the IEEE Industrial Electronics Society*. 767–772. <https://doi.org/10.1109/IECON.2016.7792958>
- [13] Koen Crommentuijn and Fredrik Winberg. 2006. Designing Auditory Displays to Facilitate Object Localization in Virtual Haptic 3D Environments. In *Proceedings of the 8th International ACM SIGACCESS Conference on Computers and Accessibility* (Portland, Oregon, USA) (Assets '06). Association for Computing Machinery, New York, NY, USA, 255–256. <https://doi.org/10.1145/1168987.1169042>
- [14] Florian Dramas, Bernard Oriola, Brian G. Katz, Simon J. Thorpe, and Christophe Jouffrais. 2008. Designing an Assistive Device for the Blind Based on Object Localization and Augmented Auditory Reality. In *Proceedings of the 10th International ACM SIGACCESS Conference on Computers and Accessibility* (Halifax, Nova Scotia, Canada) (Assets '08). Association for Computing Machinery, New York, NY, USA, 263–264. <https://doi.org/10.1145/1414471.1414529>
- [15] Scott Freundschuh and Max Egenhofer. 1997. Human conceptions of spaces: Implications for GIS. *Transactions in GIS* 2, 4 (1997), 361–375. <https://doi.org/10.1111/j.1467-9671.1997.tb00063.x>
- [16] William Grussenmeyer and Eelke Folmer. 2017. Accessible Touchscreen Technology for People with Visual Impairments: A Survey. *ACM Trans. Access. Comput.* 9, 2, Article 6 (Jan. 2017), 31 pages. <https://doi.org/10.1145/3022701>
- [17] Tiago Guerreiro, Kyle Montague, João Guerreiro, Rafael Nunes, Hugo Nicolau, and Daniel J.V. Gonçalves. 2015. Blind People Interacting with Large Touch Surfaces: Strategies for One-Handed and Two-Handed Exploration. In *Proceedings of the 2015 International Conference on Interactive Tabletops & Surfaces* (Madeira, Portugal) (ITS '15). Association for Computing Machinery, New York, NY, USA, 25–34. <https://doi.org/10.1145/2817721.2817743>
- [18] Jeremy Hill and John Black. 2003. The miniguide: a new electronic travel device. *Journal of Visual Impairment & Blindness* 97, 10 (2003), 1–6. <https://doi.org/10.1177/0145482X0309701010>
- [19] Matthias Hoppe, Pascal Knierim, Thomas Kosch, Markus Funk, Lauren Futami, Stefan Schneeegg, Niels Henze, Albrecht Schmidt, and Tonja Machulla. 2018. VRHapticDrones: Providing Haptics in Virtual Reality Through Quadcopters. In *Proceedings of the 17th International Conference on Mobile and Ubiquitous Multimedia* (Cairo, Egypt) (MUM 2018). ACM, New York, NY, USA, 7–18. <https://doi.org/10.1145/3282894.3282898>
- [20] L. A. Johnson and C. M. Higgins. 2006. A Navigation Aid for the Blind Using Tactile-Visual Sensory Substitution. In *2006 International Conference of the IEEE Engineering in Medicine and Biology Society*. 6289–6292.
- [21] Esteban Bayro Kaiser and Michael Lawo. 2012. Wearable navigation system for the visually impaired and blind people. In *2012 IEEE/ACIS 11th International Conference on Computer and Information Science*. IEEE, IEEE, Shanghai, China, 230–233. <https://doi.org/10.1109/ICIS.2012.118>
- [22] Shaun K. Kane, Meredith Ringel Morris, Annuska Z. Perkins, Daniel Wigdor, Richard E. Ladner, and Jacob O. Wobbrock. 2011. Access Overlays: Improving Non-Visual Access to Large Touch Screens for Blind Users. In *Proceedings of the 24th Annual ACM Symposium on User Interface Software and Technology* (Santa Barbara, California, USA) (UIST '11). Association for Computing Machinery, New York, NY, USA, 273–282. <https://doi.org/10.1145/2047196.2047232>
- [23] Kari Daniel Karjalainen, Anna Elisabeth Sofia Romell, Photchara Ratsamee, Asim Evren Yantac, Morten Fjeld, and Mohammad Obaid. 2017. Social Drone Companion for the Home Environment: A User-Centric Exploration. In *Proceedings of the 5th International Conference on Human Agent Interaction* (Bielefeld, Germany) (HAI '17). ACM, New York, NY, USA, 89–96. <https://doi.org/10.1145/3125739.3125774>
- [24] Brian FG Katz, Slim Kammoun, Gaëtan Parsehian, Olivier Gutierrez, Adrien Brilhault, Malika Auvray, Philippe Truillet, Michel Denis, Simon Thorpe, and Christophe Jouffrais. 2012. NAVIG: augmented reality guidance system for the visually impaired. *Virtual Reality* 16, 4 (2012), 253–269.
- [25] Vinitha Khambadkar and Eelke Folmer. 2013. GIST: A Gestural Interface for Remote Nonvisual Spatial Perception. In *Proceedings of the 26th Annual ACM*

- Symposium on User Interface Software and Technology* (St. Andrews, Scotland, United Kingdom) (UIST '13). Association for Computing Machinery, New York, NY, USA, 301–310. <https://doi.org/10.1145/2501988.2502047>
- [26] Pascal Knierim, Thomas Kosch, Alexander Achberger, and Markus Funk. 2018. Flyables: Exploring 3D Interaction Spaces for Levitating Tangibles. In *Proceedings of the Twelfth International Conference on Tangible, Embedded, and Embodied Interaction* (Stockholm, Sweden) (TEI '18). ACM, New York, NY, USA, 329–336. <https://doi.org/10.1145/3173225.3173273>
- [27] Maaïke Langelaan, Michiel R de Boer, Ruth MA van Nispen, Bill Wouters, Annette C Moll, and Ger HMB van Rens. 2007. Impact of visual impairment on quality of life: a comparison with quality of life in the general population and with other chronic conditions. *Ophthalmic epidemiology* 14, 3 (2007), 119–126. <https://doi.org/10.1080/09286580601139212>
- [28] S. J. Lederman and R. L. Klatzky. 2009. Haptic perception: A tutorial. *Attention, Perception, & Psychophysics* 71, 7 (2009), 1439–1459. <https://doi.org/10.3758/APP.71.7.1439>
- [29] Sooyeon Lee, Chien Wen Yuan, Benjamin V. Hanrahan, Mary Beth Rosson, and John M. Carroll. 2017. Reaching Out: Investigating Different Modalities to Help People with Visual Impairments Acquire Items. In *Proceedings of the 19th International ACM SIGACCESS Conference on Computers and Accessibility* (Baltimore, Maryland, USA) (ASSETS '17). Association for Computing Machinery, New York, NY, USA, 389–390. <https://doi.org/10.1145/3132525.3134817>
- [30] Xiaoping Liu, He Zhang, Lingqiu Jin, and Cang Ye. 2018. A Wearable Robotic Object Manipulation Aid for the Visually Impaired. In *2018 IEEE 1st International Conference on Micro/Nano Sensors for AI, Healthcare, and Robotics (NSENS)*. IEEE, Shenzhen, China, 5–9. <https://doi.org/10.1109/NSENS.2018.8713628>
- [31] Activated Realities Ltd. 2019. SenseAR. <https://www.activatedrealities.com/SenseAR>
- [32] S. Mashal, M. Kranz, and G. Hoelzl. 2020. Do You Feel Like Flying? A Study of Flying Perception in Virtual Reality for Future Game Development. *IEEE Computer Graphics and Applications* 40, 4 (2020), 51–61.
- [33] A. A. Nada, M. A. Fakhr, and A. F. Seddik. 2015. Assistive infrared sensor based smart stick for blind people. In *2015 Science and Information Conference (SAI)*. 1149–1154. <https://doi.org/10.1109/SAI.2015.7237289>
- [34] John Nicholson, Vladimir Kulyukin, and Daniel Coster. 2009. ShopTalk: independent blind shopping through verbal route directions and barcode scans. *The Open Rehabilitation Journal* 2, 1 (2009), 11–23. <https://doi.org/10.2174/1874943700902010011>
- [35] S. Panëels, M. Anastassova, S. Strachan, S. P. Van, S. Sivacoumarane, and C. Bolzmacher. 2013. What's around me? Multi-actuator haptic feedback on the wrist. In *2013 World Haptics Conference (WHC)*. 407–412.
- [36] Clinton R. Sanders. 2000. The Impact of Guide Dogs on the Identity of People with Visual Impairments. *Anthrozoös* 13, 3 (2000), 131–139. <https://doi.org/10.2752/089279300786999815>
- [37] S. Satpute, J. Canady, R. L. Klatzky, and G. Stetten. 2019. FingerSight: A Vibrotactile Wearable Ring for Assistance with Locating and Reaching Objects in Peripersonal Space. *IEEE Transactions on Haptics* (2019), 1–1. <https://doi.org/10.1109/TOH.2019.2945561>
- [38] Shraga Shoval, Johann Borenstein, and Yoram Koren. 1994. Mobile robot obstacle avoidance in a computerized travel aid for the blind. In *Proceedings of the 1994 IEEE International Conference on Robotics and Automation*. IEEE, IEEE, San Diego, CA, USA, 2023–2028. <https://doi.org/10.1109/ROBOT.1994.351166>
- [39] P. Skulimowski, P. Korb, and P. Wawrzyniak. 2014. POI explorer - A sonified mobile application aiding the visually impaired in urban navigation. In *2014 Federated Conference on Computer Science and Information Systems*. 969–976. <https://doi.org/10.15439/2014F293>
- [40] A. K. Srinivasan, S. Sridharan, and R. Sridhar. 2020. Object Localization and Navigation Assistant for the Visually challenged. In *2020 Fourth International Conference on Computing Methodologies and Communication (ICCMC)*. 324–328.
- [41] ArduPilot Dev Team. 2019. ARDUPILOT - Versatile, Trusted, Open. ArduPilot. Retrieved February 06, 2020 from <https://www.ardupilot.org/>
- [42] PX4 Dev Team. 2019. PX4 - Omnibus F4 SD. PX4. Retrieved February 06, 2020 from https://docs.px4.io/master/en/flight_controller/omnibus_f4_sd.html
- [43] W. M. Bergmann Tiest and A. M.L. Kappers. 2015. Haptic perception of force. *Scholarpedia* 10, 4 (2015), 32732. <https://doi.org/10.4249/scholarpedia.32732>
- [44] A. Wachaja, P. Agarwal, M. Zink, M. R. Adame, K. Möller, and W. Burgard. 2015. Navigating blind people with a smart walker. In *2015 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 6014–6019. <https://doi.org/10.1109/IROS.2015.7354233>
- [45] H. Wang, R. K. Katschmann, S. Teng, B. Araki, L. Giarré, and D. Rus. 2017. Enabling independent navigation for visually impaired people through a wearable vision-based feedback system. In *2017 IEEE International Conference on Robotics and Automation (ICRA)*. 6533–6540. <https://doi.org/10.1109/ICRA.2017.7989772>
- [46] Graham Wilson and Stephen A. Brewster. 2016. Using Dynamic Audio Feedback to Support Peripersonal Reaching in Young Visually Impaired People. In *Proceedings of the 18th International ACM SIGACCESS Conference on Computers and Accessibility* (Reno, Nevada, USA) (ASSETS '16). Association for Computing Machinery, New York, NY, USA, 209–218. <https://doi.org/10.1145/2982142.2982160>
- [47] Jeff Wilson, Bruce N. Walker, Jeffrey Lindsay, Craig Cambias, and Frank Dellaert. 2007. SWAN: System for Wearable Audio Navigation. In *Proceedings of the 2007 11th IEEE International Symposium on Wearable Computers (ISWC '07)*. IEEE Computer Society, USA, 1–8. <https://doi.org/10.1109/ISWC.2007.4373786>
- [48] Winston Wu, Lawrence Au, Brett Jordan, Thanos Stathopoulos, Maxim Batalin, William Kaiser, Alireza Vahdatpour, Majid Sarrafzadeh, Meika Fang, and Joshua Chodosh. 2008. The smartcane system: an assistive device for geriatrics. In *Proceedings of the ICST 3rd international conference on Body area networks*. ICST, ACM, Tempe, Arizona, 2. <https://doi.org/10.4108/icst.bodynets2008.2944>